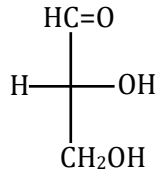
RACE
Regular Analysis Through Continuous Exercise

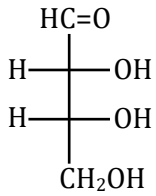
Biomolecules : RACE-01

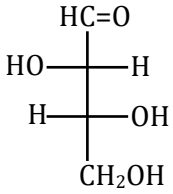
1. Glucose is a/an :

(A) Aldohexose (B) Aldopentose (C) Aldotetrose (D) Ketohexose

2. Which of the following carbohydrate has D configuration ?

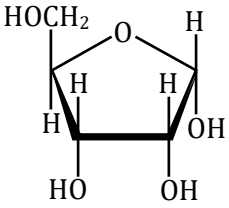
(A) 

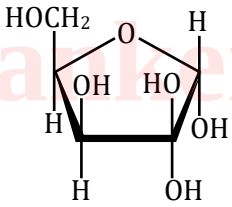
(B) 

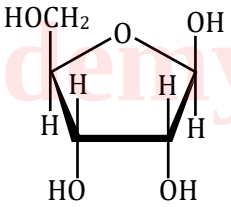
(C) 

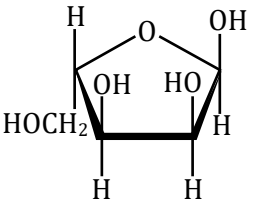
(D) All of these

3. Which of the following represents the anomer of the compound shown ?



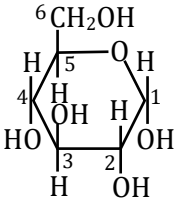
(A) 

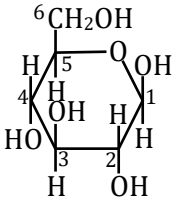
(B) 

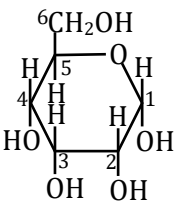
(C) 

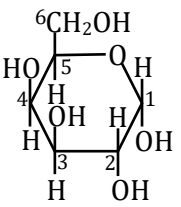
(D) None of these

4. Which of the following is the structure of α-D-(+) Glucopyranose

(A) 

(B) 

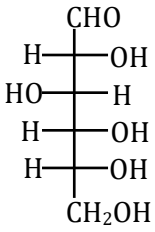
(C) 

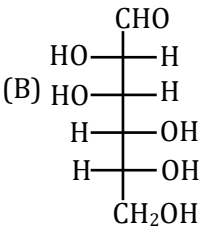
(D) 

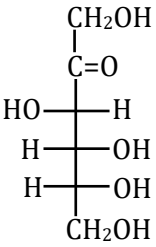
Pankers Academy JEE

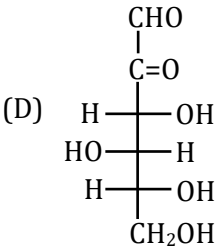
For More Join: @IITJEE_Advanced

5. When D-Glucose is placed in basic aqueous solution an equilibrium mixture of three compounds is obtained. Which of the following will not be present in equilibrium mixture ?

(A) 

(B) 

(C) 

(D) 

6. Among the following, the incorrect statement is :
(A) Cellulose and amylose have 1, 4-glycosidic linkage.
(B) Lactose contains β-D-galactose and β-D-glucose.
(C) Maltose and lactose have 1, 4-glycosidic linkage.
(D) Sucrose and amylose have 1, 2-glycosidic linkage.

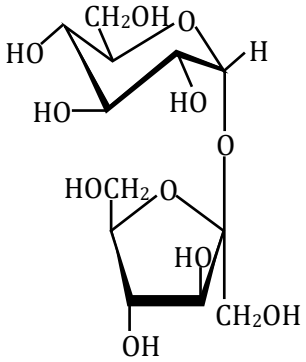
7. Which of the following tests is given by fructose?
(A) Biuret test (B) Xanthoproteic test
(C) Barfoed test (D)Ninhydrin test

8. Fructose and glucose can be distinguished by:
(A) Tollen's test (B) $Br_2 + H_2O$ (C) Fehling's test (D) Benedict test

9. Which one of the following pairs is correctly matched?
(A) Sucrose : reducing sugar (B) Glucose : mutarotation
(C) Fructose : aldose sugar (D) Sucrose : monosaccharide

10. What is incorrect about cellulose ?
(A) It is a polysaccharide (B) It has β–1,4'-glycosidic bond
(C) One of the hydrolysis product is galactose (D) It is non reducing sugar

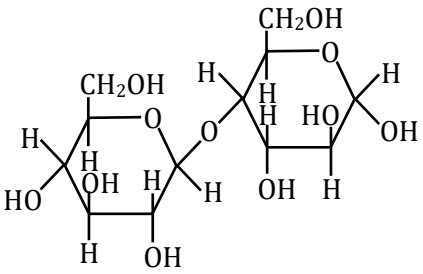
11. What is incorrect about the given structure ?



(A) It is sucrose
(B) It is a reducing sugar
(C) It has α-C_1 glucopyranoside and β-C_2 fructofuranoside bond
(D) On hydrolysis it gives glucose and fructose

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12. The correct statement about the following sugars X is:



(X)

(A) X is a non-reducing sugar

(B) X does not have hemiacetal

(C) The glucosidic linkages in X is α-glycoside

(D) The glucosidic linkages in X is β-glycoside

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	D	B	A	D	D	C	B	B	C
Question	11	12								
Answer	B	D								

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RACE
Regular Analysis Through Continuous Exercise

Biomolecules : RACE-02

1. Which of the following is essential amino acid?
(A) Glutamic acid (B) Glycine (C) Phenylalanine (D) Aspartic acid

2. Which of the following is basic amino acid?
(A) Glycine (B) Valine (C) Histidine (D) Alanine

3. An amino acid has pI = 10.8. It may be
(A) Acidic amino acid (B) Basic amino acid
(C) Neutral amino acid (D) None of these

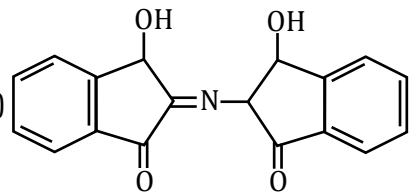
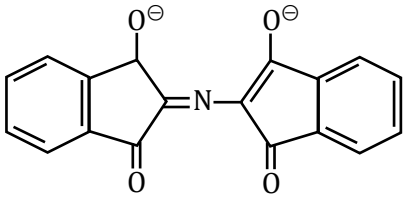
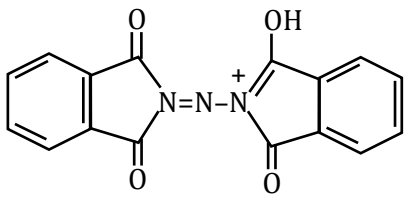
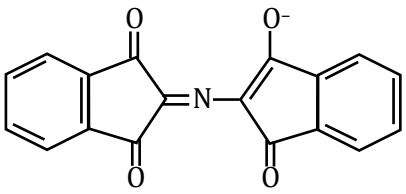
4. The three pK_a values of aspartic acid are 1.89, 3.65 and 9.60. The pI of the amino acid is:
(A) 2.77 (B) 6.62 (C) 5.74 (D) 7.0

5. At iso-electric point:
(A) Concentration of cation is equal to concentration of anion
(B) Net charge is zero
(C) Maximum conc. of di-pole ion (Zwitter ion) will be present
(D) All of the above

6. Peptide linkage is:
(A) $-\overset{O}{\overset{\|}{C}}-O-$ (B) $-\overset{O}{\overset{\|}{C}}-NH_2$ (C) $-\overset{O}{\overset{\|}{C}}-NH-$ (D) $-\overset{O}{\overset{\|}{C}}-NH-NH_2$

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7. The correct structure of purple coloured compound formed in the reaction of ninhydrin with proteins is:

(A) 
(B) 
(C) 
(D) 

8. The secondary structure of protein is stabilised by:
(A) Peptide bond (B) Glycosidic bond (C) Hydrogen bonding (D)Vander Waals forces

9. The presence or absence of hydroxyl group on which carbon atom of sugar differentiates RNA and DNA ?
(A) 1st (B) 2nd (C) 3rd (D) 4th

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10. Sugar moiety in DNA and RNA molecules respectively are

(A) β-D-2-deoxyribose, β-D-deoxyribose (B) β-D-2-deoxyribose, β-D-ribose

(C) β-D-ribose, β-D-2-deoxyribose (D) β-D-deoxyribose, β-D-2-deoxyribose

11. When adenine is attached to ribose sugar, it is called adenosine. To make a nucleotide from it, it would require

(A) Oxygenation (B) addition of a base

(C) addition of phosphate (D) hydrogenation

12. The function of DNA in an organism is

(A) to assist in the synthesis of RNA molecule

(B) to store information of heredity characteristics

(C) to assist in the synthesis of proteins and polypeptides

(D) All of these

13. In both DNA and RNA, heterocyclic base and phosphate ester linkages are at-

(A) C'_5 and C'_1 respectively of the sugar molecule

(B) C'_1 and C'_5 respectively of the sugar molecule

(C) C'_2 and C'_5 respectively of the sugar molecule

(D) C'_5 and C'_2 respectively of the sugar molecule

14. Bone deformities in children is due to the deficiency of :

(A) Vitamin A (B) Vitamin B_6

(C) Vitamin D (D) Vitamin C

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	C	B	A	D	C	D	C	B	B
Question	11	12	13	14						
Answer	C	D	B	C						

For More Join: @IITJEE_Advanced